Humans and transformer LMs: Abstraction drives language learning

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Best Paper Award at EACL 2026

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Abstract

Categorization is a core component of human linguistic competence. We investigate how a transformer-based language model (LM) learns linguistic categories by comparing its behaviour over the course of training to behaviours which characterize **abstract feature**–based and **concrete exemplar**–based accounts of human language acquisition. We investigate how lexical semantic and syntactic categories emerge using novel divergence-based metrics that track learning trajectories using next-token distributions. In experiments with GPT-2 small, we find that (i) when a construction is learned, abstract class-level behaviour is evident at earlier steps than lexical item–specific behaviour, and (ii) that different linguistic behaviours emerge abruptly in sequence

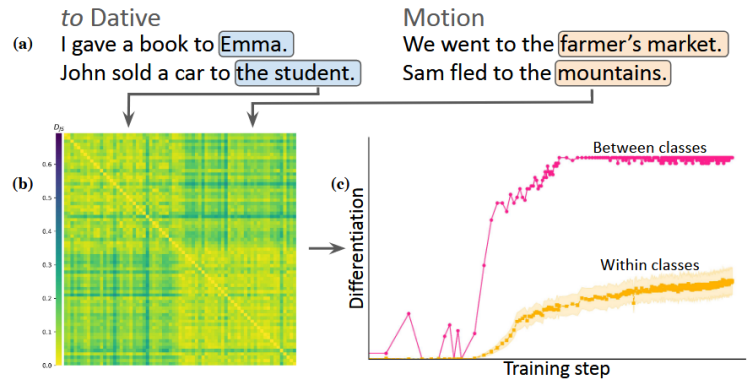


Figure 1: (a) Categories, like verb classes determined by argument structure, are pervasive in language: *to* Dative verbs take recipient arguments, whereas Motion verbs take goal locations. (b) Divergence metrics compare LM prediction distributions conditioned on different categories. (c) Tracking divergences over training reveals that categories like verb classes are differentiated early by LMs, before gradual item-level learning.

I. Introduction

- Investigate how a transformer-based language models (LM) **learn linguistic categories** by comparing its behavior over the course of training
- **Abstraction-first account:** linking the input data to abstract featural representations
- **Exemplar-first account:** storing linguistic examples observed in the input

II. What drive argument structure learning in LMs?

- Measures the arguments predicted for each verb as a proxy for verb categorization.

CLASS	EXAMPLE VERBS	EXAMPLE SENTENCES	TOTAL
(i) *to* Dative	*give*, *sell*, *grant* (*n*=35)	*Chipotle* ***gave*** *away free burritos to the* __	1841
(ii) Motion	*go*, *walk*, *flee* (*n*=36)	*Toby accordingly* ***goes*** *to the* __	1942
(iii) Reciprocal	*speak*, *meet*, *talk* (*n*=16)	*Cherry recalled Orr had refused to* ***speak*** *with the* __	465
(iv) *spray-load*	*spray*, *load*, *smear* (*n*=16)	*Ray and Devon then* ***sprayed*** *the table with the* __	404

Table 1: Example verbs and sentence prefixes from the four verb classes used in the experiments.

Class	Argument
(i) *to* Dative	recipients
(ii) Motion	goal location
(iii) Reciprocal	reciprocal object
(iv) *spray-load*	substances

II. What drive argument structure learning in LMs?

- Examine next-token distributions
- Track whether predictions change when **comparing any two verb** (evidence for **item-specific learning**) **or predictions are more similar for verbs that belong to the same class than for verbs that belong to a separate comparison class** (evidence for **class-general learning**)

For each verb ***v***, this is defined as the arithmetic **mean of the conditional next-token distributions** over the set $\mathbf{S_v}$ of **N** prefixes containing verb ***v***:

$$P_v(x) = \frac{1}{N}\sum_{i=1}^{N} P(x|s_i),\ s_i \in S_v.$$

- Compare similarity scores between verbs by taking the **Jensen-Shannon divergence**, $\mathbf{D_{js}}$ between the distribution defined above

II. What drive argument structure learning in LMs?

- **Item learning metric:** measure the emergence of **item-specific learning independent of effect of verb class** as the mean $\mathbf{D_{js}}$ between verb ***v*** and verbs within the same class. If in-class $\mathbf{D_{js}}$ increases, then idiosyncratic properties of verb ***v*** that are not general to the class have been learned.
- **Class learning metric:** measure $\mathbf{D_{js}}$ between verb ***v*** with verb from the same class **UA** and verb from different class **UB**

Model: **training checkpoints of GPT-2,** tracking behavior over 450 check points (400K step)

II. What drive argument structure learning in LMs?

- For the **(i) *to* Dative** and **(ii) Motion class**, **class-based distinction** emerged 50 steps before **in-class difference** begin

- For the **(iii) Reciprocal** and **(iv) spray-load class**, the onset of change is **concurrent**

Figure 3: Mann-Whitney U-test between $U_A = \{D_{JS}(P_{v_t}(x)||P_v(x)) : v, v_t \in C_1\}$ (same class), $U_B = \{D_{JS}(P_{v_t}(x)||P_v(x)) : v_t \in C_1, v \in C_2\}$ (different class), for verb classes C_1, C_2. Mean of D_{JS} across verbs in the same class. Shading is standard deviation.

II. What drive argument structure learning in LMs?

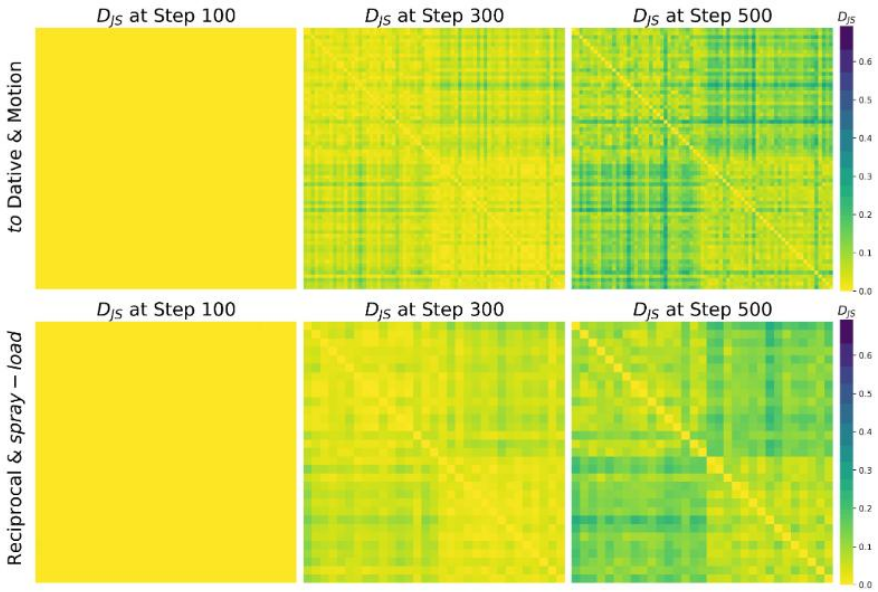
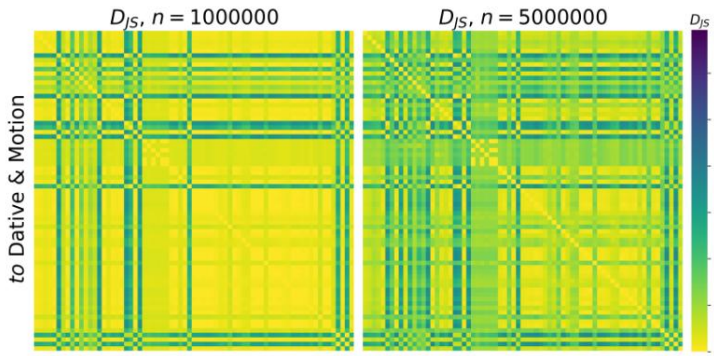


Figure 2: Pairwise D_{JS} between verbs in two classes at three early steps in training. Each row and column represents a single verb. Differentiation into verb classes is evident when any change has occurred – between-class quadrants have higher D_{JS} (top-right, bottom-left) than in-class quadrants (top-left, bottom-right).

II. What drive argument structure learning in LMs?

- **Exemplar-first baseline:** co-occurrence statistics for each verb in the correct prepositional structure, using OpenWebText corpus



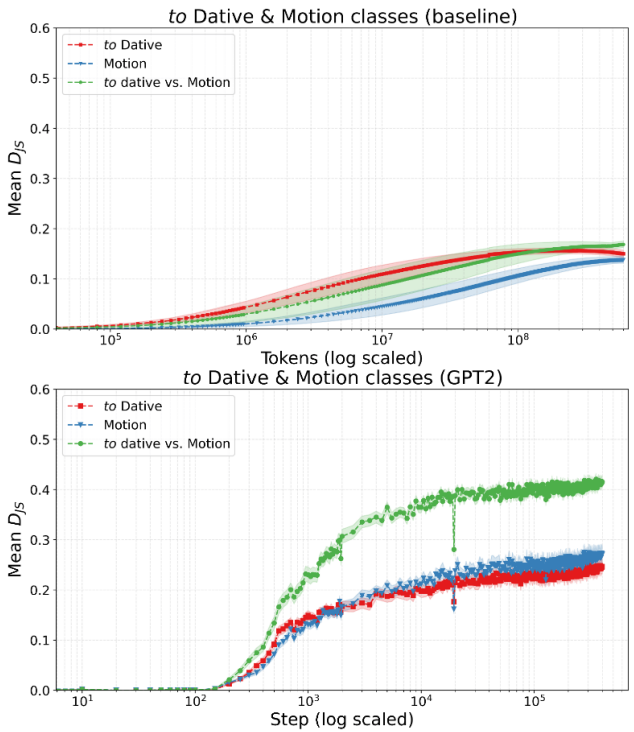


Figure 5: **Top:** Pairwise D_{JS} across two classes in exemplar-first baseline show verb-specific patterns emerge without clear class distinctions. **Middle and bottom:** Exemplar-first baseline (middle) shows increasing divergence between verbs driven by verb-specific learning, before decrease, while GPT-2 is monotonic (bottom); baseline between-class mean divergences are not consistently greater than in-class means, unlike GPT-2. Shading is 95% CI.

- In the exemplar-first baseline, verbs in the to Dative class diverge before converge again when more data is observed
- In GPT-2, these classes are differentiated early and consistently throughout training

→ do not match exemplar-first accounts

II. Is learning generally abstraction-driven?

- If constructions are learned **in tandem for classes of lexical items**, then we can say that LMs are corresponding with an **abstraction-first** account.

- If they **learn constructions idiosyncratically for individual instances, before gradually generalizing across classes**, then LM learning better corresponds with an **exemplar-first account.**

STRUCTURE	EXAMPLE SENTENCES
TRANSITIVITY (n=1000)	The truck hadn't **astounded** __ The truck hadn't **fallen** __
RELATIVE CLAUSE (*to* Dative; n=824)	The **person** that they sold the land to __ The person **thinks** that they sold the land to __
RELATIVE CLAUSE (Reciprocal; n=288)	The **person** that Optimus spoke with __ The person **thinks** that Optimus spoke with __

Table 2: Example structures for Experiment 2. TRANSITIVITY is from BLiMP (Warstadt et al., 2020).

II. Is learning generally abstraction-driven?

Two types of structures were studied

- **TRANSITIVITY**: Verbs differ in whether or not they require an object: *transitive* and *intransitive* categorization
- **RELATIVE CLAUSE**: The two categories verbgap and verbno gap contain the same verbs in different syntactic context.
  {$give_{\text{no gap}}$, $sell_{\text{no gap}}$} and {$give_{\text{gap}}$, $sell_{\text{gap}}$}

(3) *The person that* ... VERB_{GAP} ... __

(4) *The person thinks that* ... $\text{VERB}_{\text{NO GAP}}$ ... __

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II. Is learning generally abstraction-driven?

- An increase in pairwise $\mathbf{D_{js}}$ between **minimal pairs always occurs with increases in verbs which are systematically more similar to those belonging to the same category, than those which do not**

→ Item-based and class-based learning occur almost concurrently

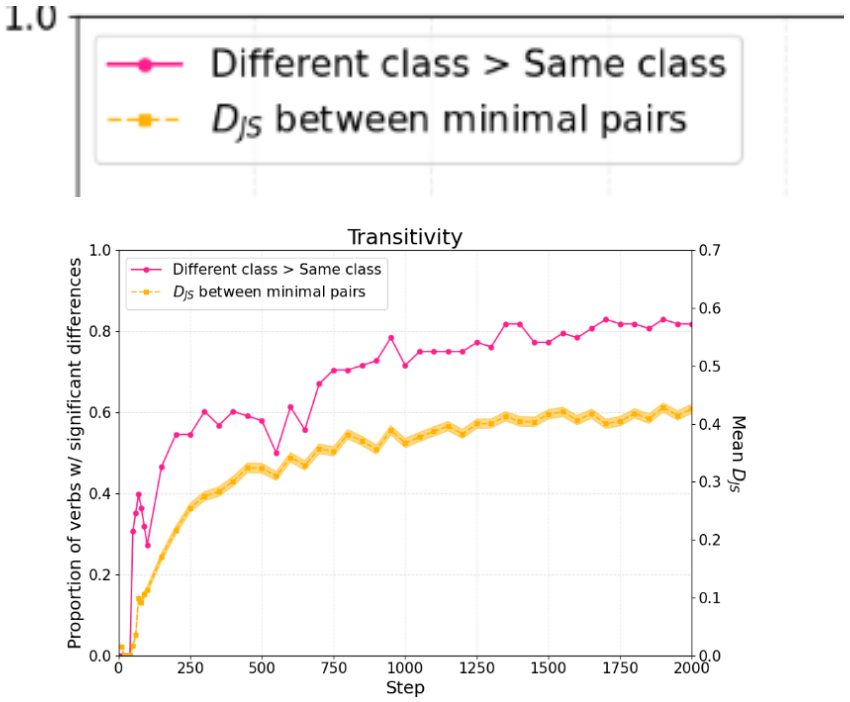

Figure 6: Comparing class-level and item-based behaviour for TRANSITIVITY dataset.

II. Is learning generally abstraction-driven?

- **Structures are not learned globally**

Relative clause filler-gap categories are not learned for all verbs at once. They are learned separately for each class

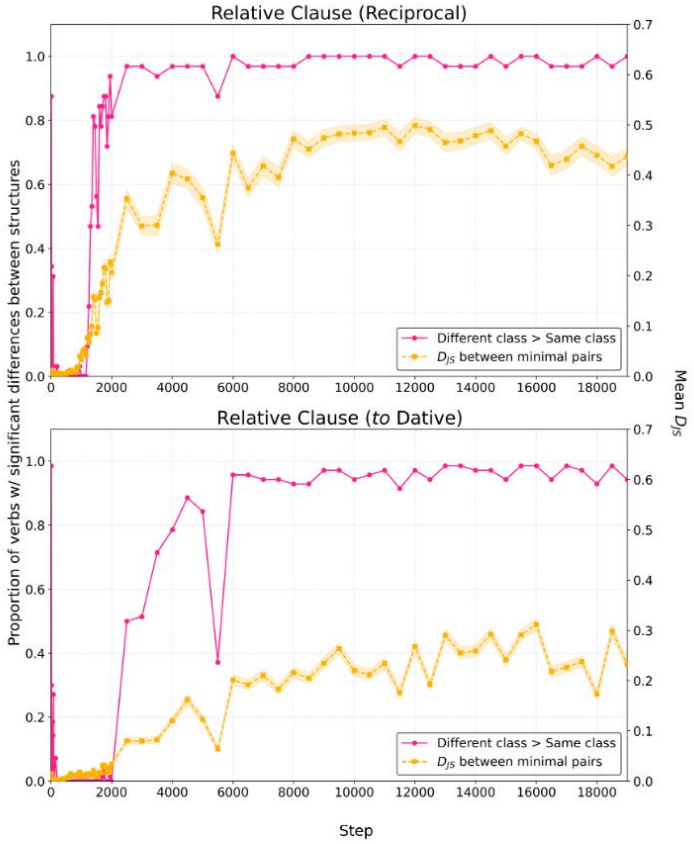

Figure 7: Comparing class-level and item-based behaviour for RELATIVE CLAUSE datasets.

II. Is learning generally abstraction-driven?

- **Structures are learned sequentially**

Each of the abstractions emerges with distinct onset in the same sequence over the course of training.

Syntactic subcategorization
(TRANSITIVITY)
↓
Argument structure
(VERB CLASS)
↓
Non-local dependencies
(RELATIVE CLAUSE)

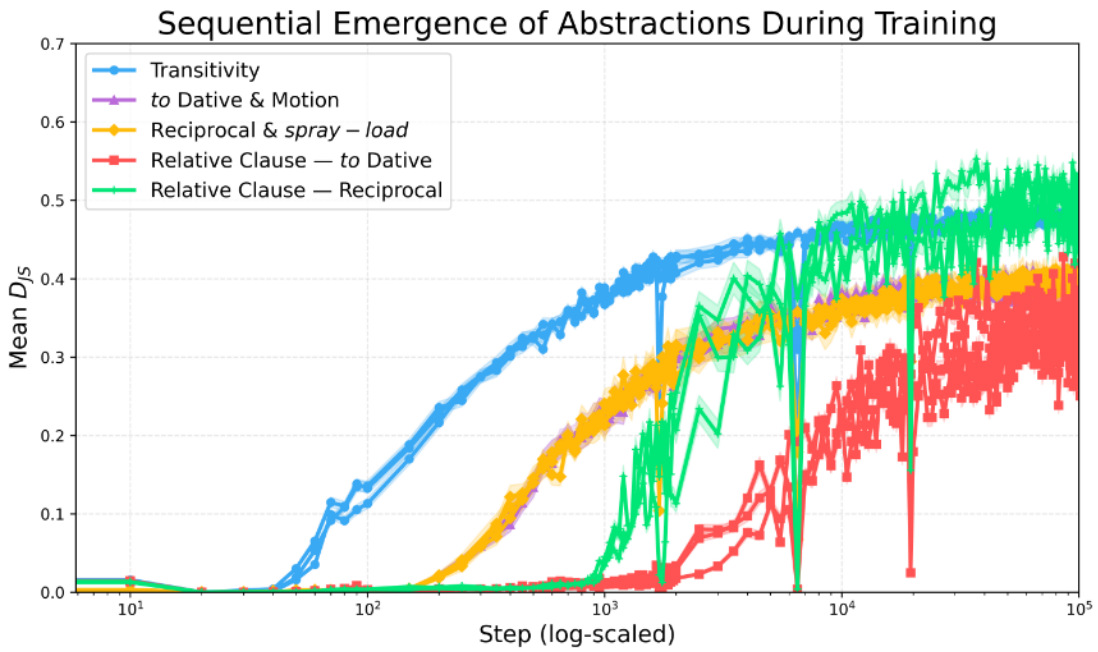

Figure 8: Learning curves for each abstraction plotted for three models trained with different random seeds.